

GRADE 7 · READING INFORMATIONAL TEXT · RI.7.3

Interactions in Informational Text

Analyze how individuals, events, and ideas interact to produce the development of the polio vaccine.

Lesson Map · Passage · Guided · Partner · Independent Practice · Small Group · Answer Key

What You Will Learn Today

Learning Goals

Understand RI.7.3: Analyze how individuals, events, and ideas interact to produce the dev...

Read an original passage closely

Support every answer with text evidence

Work through guided, partner, and independent practice

How the Lesson Runs

Warm-up and passage

Guided practice together

Partner discussion

Independent + small-group work

Exit ticket and answer key

Unpacking the Standard

Informational texts about history and science often show individuals, events, and ideas affecting each other, not existing separately. An idea can push a person to act; a person's action can create an event; an event can change how people think about an idea.

Modeled Example

An outbreak (event) causes a scientist (individual) to change her research focus (idea).
What kind of interaction is this?

An event influencing an individual, which then changes the direction of an idea (her research focus).

A new theory (idea) convinces a government to fund a research program (event). What kind of interaction is this?

An idea influencing an event, since the theory directly led to a concrete action (funding the program).

Racing an Invisible Enemy

1. In the summer of 1952, the United States recorded its worst polio outbreak in history: nearly 58,000 cases, many resulting in paralysis, some in death. Public pools closed. Parents kept children indoors. The virus, which attacked the nervous system, could not be reliably treated, only prevented — and no one had yet developed a way to prevent it.
2. Jonas Salk, a virologist at the University of Pittsburgh, had spent the previous several years developing a vaccine using an unconventional approach: a "killed-virus" method, using inactivated poliovirus that could not cause infection but might still train the immune system to recognize it.
3. Many of Salk's colleagues favored a competing idea: a "live-virus" vaccine, using a weakened but still-living form of the virus, which some researchers believed would produce stronger, longer-lasting immunity. The scientific community was divided, and the 1952 outbreak added urgency to a debate that had, until then, moved slowly.
4. The scale of the outbreak influenced funding decisions directly. The National Foundation for Infantile Paralysis, already collecting small donations from millions of Americans through its March of Dimes campaign, redirected significant resources toward accelerating vaccine trials.

Racing an Invisible Enemy, continued

5. In 1954, backed by that funding, Salk's killed-virus vaccine entered a massive field trial — nearly two million children participated, making it one of the largest medical experiments in history to that point. The scale of the outbreak had made the scale of the trial possible; without the public fear and urgency 1952 created, funding on that level would have been far harder to secure.

6. The trial's results, announced in April 1955, showed the vaccine was safe and effective. The announcement triggered a new event: a national rollout so urgent that some manufacturers rushed production, leading to a separate, serious problem when one improperly prepared batch caused a small number of actual polio cases before stricter safety protocols were enforced.

7. Cases in the United States fell from tens of thousands per year in the early 1950s to under a thousand by the early 1960s. The idea that had once divided the scientific community — a killed, inactivated virus could safely produce immunity — was, by then, no longer in question.

Warm-up · Before You Read

1. What individual is introduced in paragraph 2, and what idea is he working on?

2. What event does paragraph 1 describe, and what were its effects?

Guided Practice

3. How did the event in paragraph 1 interact with the idea described in paragraph 3?

4. What competing idea does paragraph 3 introduce, and who favored it?

Guided Practice

5. How did the event (the outbreak) influence funding, according to paragraph 4?

6. How did that funding (influenced by the event) affect Salk's work, according to paragraph 5?

Partner Discussion

7. Explain the chain of interaction from the 1952 outbreak to the 1954 trial, using evidence from at least two paragraphs.

8. What event does paragraph 6 describe as resulting from the trial's success?

Partner Discussion

9. How did the event in paragraph 6 (the rushed rollout) create a new problem?

Independent Practice

10. How did that problem likely influence future individuals or events, based on the text's wording?

11. How did Salk's idea (paragraph 2) ultimately influence public health outcomes, according to paragraph 7?

Independent Practice

12. How does paragraph 7 show an idea changing because of events?

13. Identify one place in the text where an individual's idea directly caused an event.

Independent Practice

14. Identify one place in the text where an event directly caused a change in an idea's acceptance.

15. How might the outcome have been different if the 1952 outbreak had not occurred, based on the text's cause-and-effect pattern?

Small-Group Work

16. Explain how the March of Dimes campaign (paragraph 4) connects an event, an individual effort, and an idea.

17. Why does the text mention the competing live-virus idea in paragraph 3 instead of only describing Salk's approach?

Small-Group Work

18. How do individuals, events, and ideas each play a distinct role in this passage? Give one example of each.

19. Trace one idea across the entire passage, from its introduction to its final outcome.

Exit Ticket

20. Explain, using two pieces of evidence, how an event and an idea interacted to produce a specific outcome in this text.

Answer Key (1 of 4)

1. Jonas Salk, a virologist developing a "killed-virus" vaccine using inactivated poliovirus.
2. The 1952 polio outbreak, which caused nearly 58,000 cases and led to closed pools and parents keeping children indoors.
3. The outbreak added urgency to the scientific debate between the killed-virus and live-virus approaches, a debate paragraph 3 says had "moved slowly" before 1952.
4. A "live-virus" vaccine using a weakened but living form of the virus, favored by many of Salk's colleagues.
5. The National Foundation for Infantile Paralysis redirected significant resources from its March of Dimes campaign toward accelerating vaccine trials.

Answer Key (2 of 4)

6. It allowed his killed-virus vaccine to enter a massive 1954 field trial involving nearly two million children.

7. The 1952 outbreak (event) created urgency and fear, which led the Foundation to redirect funding (paragraph 4), and that funding then made the large-scale 1954 trial possible (paragraph 5) — an event influencing funding, which then enabled a further event.

8. A national rollout of the vaccine, announced after the trial's results were released in April 1955.

9. Some manufacturers rushed production, and one improperly prepared batch caused a small number of actual polio cases before stricter safety protocols were enforced.

10. It likely influenced regulators or manufacturers to enforce "stricter safety protocols," suggesting the problem directly caused a change in vaccine production oversight.

Answer Key (3 of 4)

11. Cases fell from tens of thousands per year in the early 1950s to under a thousand by the early 1960s, showing the vaccine's idea translated into a major real-world effect.

12. It states the once-divided scientific community no longer questioned "the idea that... a killed, inactivated virus could safely produce immunity," showing the idea gained acceptance because of the trial's results and the vaccine's real-world success.

13. Salk's idea for a killed-virus vaccine (paragraph 2) directly led to the 1954 field trial (paragraph 5), an event testing that idea.

14. The 1955 trial results (event) caused the killed-virus idea to become widely accepted, ending the earlier scientific debate described in paragraph 3.

15. Funding might not have been redirected as urgently (paragraph 4), which could have delayed or shrunk the 1954 trial, since the text ties the trial's scale directly to the outbreak's urgency.

Answer Key (4 of 4)

16. The campaign was an ongoing public effort (individuals donating) that became redirected because of the outbreak (event) to fund Salk's vaccine idea, showing all three elements working together.

17. It shows that Salk's idea was not the only option being considered, which helps explain why the urgency from the 1952 outbreak (forcing a faster decision) mattered so much to the scientific debate.

18. Individuals: Jonas Salk developing the vaccine. Events: the 1952 outbreak and the 1954 trial. Ideas: the killed-virus versus live-virus vaccine approaches.

19. The killed-virus idea is introduced in paragraph 2, debated in paragraph 3, tested at scale in paragraph 5, proven effective in paragraph 6, and fully accepted by paragraph 7 — moving from a contested idea to an established fact.

20. Answers vary; require two specific, correctly cited details showing a clear cause-and-effect interaction between an event and an idea (or individual).

Exit Ticket

Answer on a sticky note

Answer the exit-ticket question from today's lesson.

Cite at least one piece of evidence from the passage.

Remember

Name your claim or answer

Quote or cite the evidence

Explain how it supports your answer